Lecture 5: Monte Carlo Methods

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Announcements

- Reading: Chapter 10 and 11

$$E[\theta|y] \approx \frac{1}{S}\sum \theta^{(s)} \quad \text{(LLN)}$$

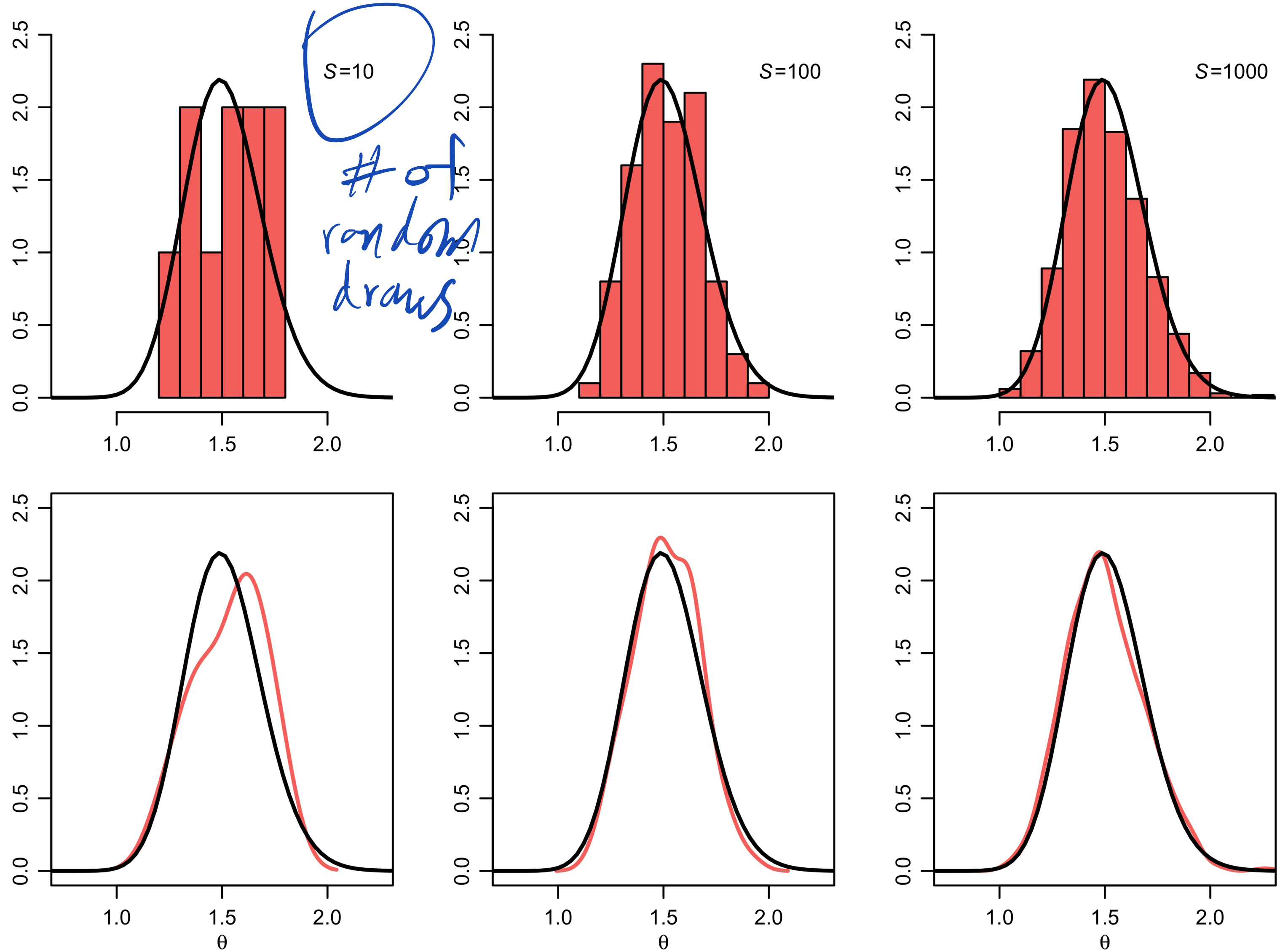
$$Var\left(\frac{1}{S}\sum \theta^{(s)}\right) = \frac{1}{S^2}\sum Var(\theta|y)$$

$$\boxed{\theta^{(s)} \text{ iid from } p(\theta|y)} \quad = \frac{Var(\theta|y)}{S}$$

<u>Monte Carlo</u> to avoid solving hard integrals.

Posterior means, credible intervals, etc.

Monte Carlo approximations of a distribution



Sampling strategies

How to get iid draws from $P(\theta | y)$

- Inversion Sampling (works for univariate distns)
- Grid sampling (works for low dimensional problems)
- Rejection sampling
- Importance sampling ← Weighted Averages
- Markov Chain Monte Carlo

Dependent Draws

Sampling strategies

Why sampling?

Probability Integral Transform

- Suppose that a random variable, Y has a continuous distribution for with CDF is F_Y.
- Then the random variable $U = F_Y(Y)$ has a uniform distribution
  - This is known as the “probability integral transform PIT”
- By taking the inverse of F_Y we have $F^{-1}(U) = Y$

Inversion Sampling

The inverse transform sampling method works as follows:

1. Generate a random number u from $\mathrm{Unif}[0, 1]$
2. Find the inverse of the desired CDF, e.g. $F_Y^{-1}(u)$.
3. Compute $y = F_Y^{-1}(u)$. y is now a sample from the desired distribution.

Inversion Sampling

Animation

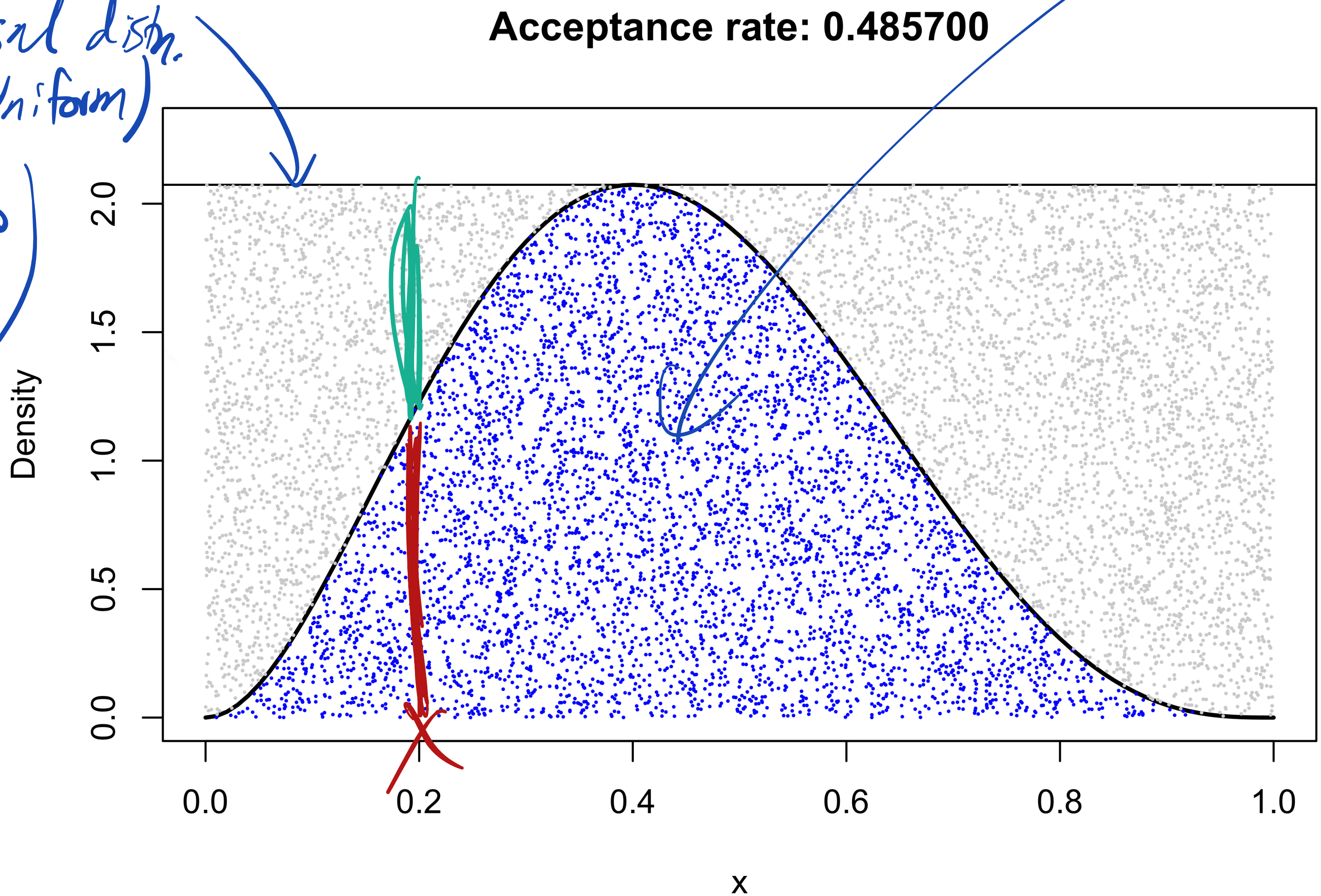
Inversion Sampling

- Inversion sampling can be a fast and simple way to sample from a distribution
- Only effective if we know the inverse-CDF and can easily compute it
- This is a big challenge in practice. For example, even the normal distribution has a CDF, Φ, which cannot be expressed analytically.
    - Shifts from one hard problem (sampling) to another (computing an integral)
    - Need alternatives!

Rejection Sampling

Sample from
a proposal distn.
(e.g. Uniform)
(easy to
sample)

Beta.



Rejection Sampling algorithm

1. Choose a proposal density, $q(\theta)$ that we can easily sample from (e.g. uniform or normal) such that:

2.Find $M = \max \frac{p(\theta|y)}{q(\theta)}$

- If $M = \infty$ then q cannot be used as a proposal distribution
- If M is finite, $Mq(\theta)$ “envelopes” $p(\theta \ y)$

3. Draw a sample, $\theta^{(s)}$ from $q(\theta)$

4. Accept $\theta^{(s)}$ as a draw from $p(\theta \mid y)$ with probability $\frac{p(\theta^{(s)}|y)}{Mq(\theta^{(s)})}$

Rejection Sampling

Demo

Rejection Sampling

- What is the expected number of rejections?
- How can we show this algorithm is valid?

Let Z be an random indicator for accepting θ^* from $q(\theta)$ targeting $P(\theta|y)$

Expected # of accepts:

$$P(Z=1) = E_q[Z] = E_q\Big[\underbrace{E[Z|\theta^*]}_{\text{Prob of accepting } \theta^*}\Big]$$

$$= E_q\left[\frac{P(\theta^*|y)}{M q(\theta^*)}\right] = \int \frac{P(\theta^*|y)}{M \cancel{q(\theta^*)}} \cancel{q(\theta^*)}\, d\theta^*$$

$$= \frac{1}{M} \int p(\theta^*|y)\, d\theta^* = \frac{1}{M}$$

Reminder: $M = \max_{\theta} \frac{p(\theta|y)}{q(\theta)}$

Validity

Have: $\tilde{p}(\theta|y) = L(\theta)p(\theta)$ (unormalized)

Want: iid Draws from $p(\theta|y)$

$$\tilde{M} = \max \tilde{p}/q$$

$\theta^* \sim q$, consider the pair (z, θ^*)

Bayes Rule to get $p(\theta^*|z=1)$ and show this equals $p(\theta|y)$

$$p(z|\theta^*) \sim \mathrm{Bern}\left(\frac{\tilde{p}}{\tilde{m}q}\right) \quad \text{(Rejection Sampling Rule)}$$

$$p(z, \theta^*) = p(\theta^*)p(z|\theta^*) =$$

$$q(\theta^*)\left(\frac{\tilde{p}(\theta^*)}{\tilde{m}q(\theta^*)}\right)^z \left(1 - \frac{\tilde{p}(\theta^*)}{\tilde{m}q(\theta^*)}\right)^{1-z}$$

$$P(\theta^* \mid z=1) = \frac{P(z=1, \theta^*)}{P(z=1)} = \frac{\cancel{q(\theta)} \dfrac{\tilde{p}(\theta \mid y)}{\cancel{\tilde{m}} \cancel{q(\theta)}}}{K / \cancel{\tilde{m}}}$$

Define $K = \int \tilde{p}(\theta \mid y)\, d\theta$ (normalizing const)

$$\tilde{m} = \max \frac{\tilde{p}}{q} = \frac{\cancel{p/K}}{q}$$

$$\tilde{m} = M / K$$

$$E\left[\frac{\tilde{p}}{\tilde{m} q}\right] = \frac{K}{\tilde{m}} = \frac{1}{M}$$

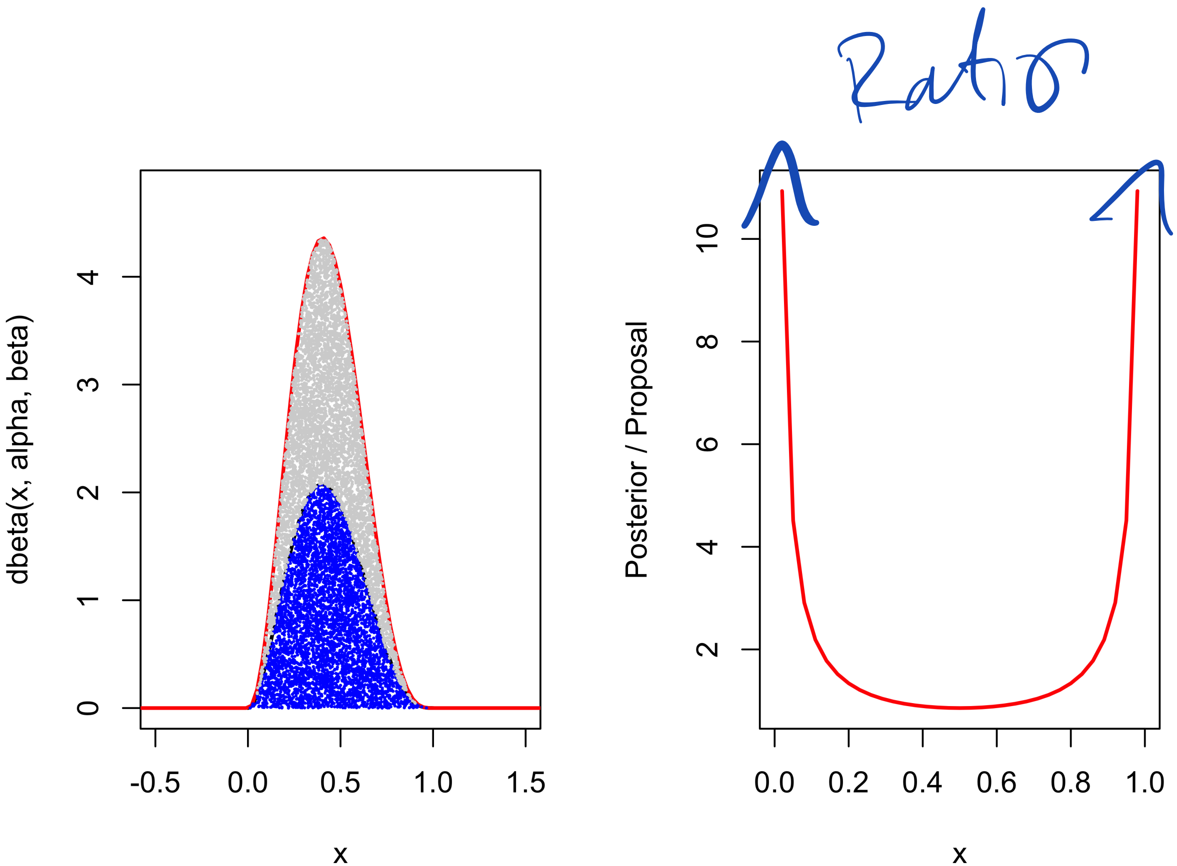
$P(z=1)$

$$= \frac{\tilde{p}(\theta \mid y)}{K}$$

$$= p(\theta \mid y)$$

Proposal most envelope target

Sometimes its not obvious…



Target density in black, proposal density in red

The German Tank Problem

The German Tank Problem

- US and allies wanted to estimate how many tanks were produced by the Germans during WWII
- They used the serial numbers on captured or destroyed tanks to estimate tank production
- Assuming serial numbers, Z_i, are ordered from 1 to n, where n is the number of tanks produced
- How do we estimate n?

The German Tank Problem

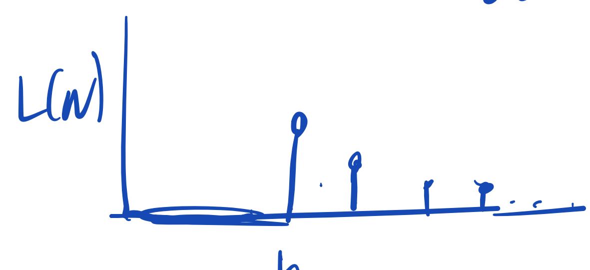
- Assume we observe the serial number of only one tank, Z
- What is a plausible sampling model? How do we write the likelihood?
- What is a reasonable prior?

$y \sim$ Discrete Unif $1, \ldots, N$.

$P(N \mid y)$?? MLE for N?

$\hat{N}_{MLE} = y$

$L(N) = I[N \geq y] \frac{1}{N}$



The German Tank Problem

- Assume Z is uniformly distributed between 1 and n
- Specify a Poisson prior distribution
  - Assume a priori a rate of 200 tanks / month
- Use Monte Carlo (rejection sampler) to summarize the results. What is a reasonable proposal?

$$P(N) \sim \frac{e^{-\lambda} \lambda^{N}}{N!}$$

$$P(N \mid y) = I[N \geq y] \frac{1}{N} e^{-\lambda} \frac{\lambda^{N}}{N!}$$

Propose from $q \sim \text{Pois}(\lambda)$

$$\tilde{m} = \max_N \frac{\tilde{P}(N|y)}{q(N)} = \frac{\frac{1}{N} \cancel{e^{-\lambda}} \cancel{\lambda^N} / \cancel{N!} \, \mathbb{I}[\ ]}{\frac{\cancel{e^{-\lambda}} \lambda^N}{\cancel{N!}}}$$

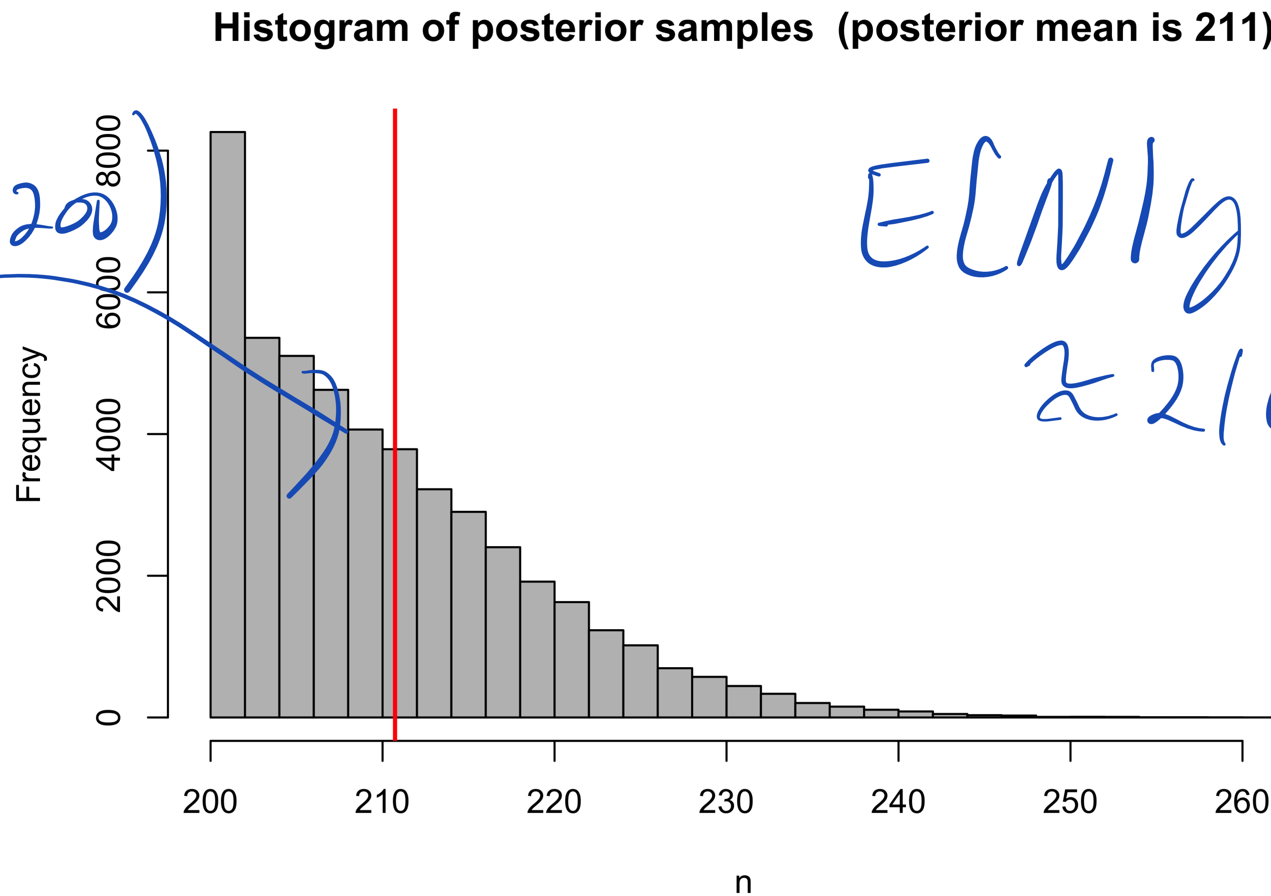
$$= \max_N \frac{1}{N} \mathbb{I}[N \geq y] = \frac{1}{y}$$

Acceptance: $$\frac{\tilde{P}(N|y)}{\tilde{m}\, q(N)} = \frac{y}{N} \mathbb{I}[N \geq y]$$

N!

The German Tank Problem, ~~Z~~ y = 200

$P(N \mid y = 200)$

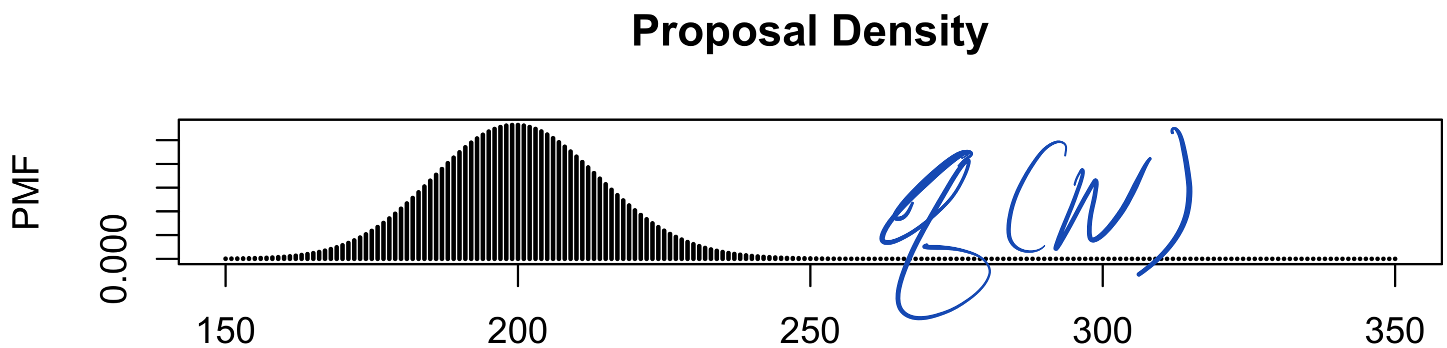


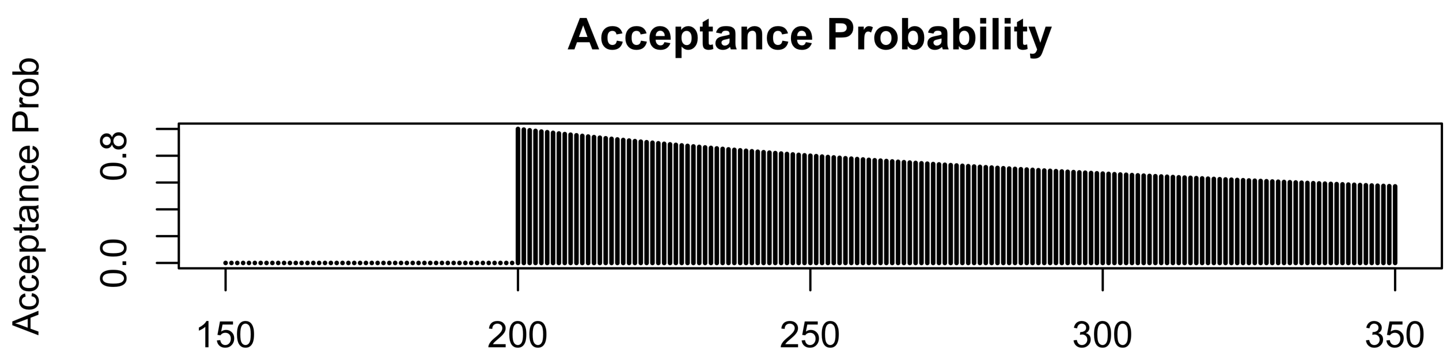
$E[N \mid y = 200]$

≈ 210

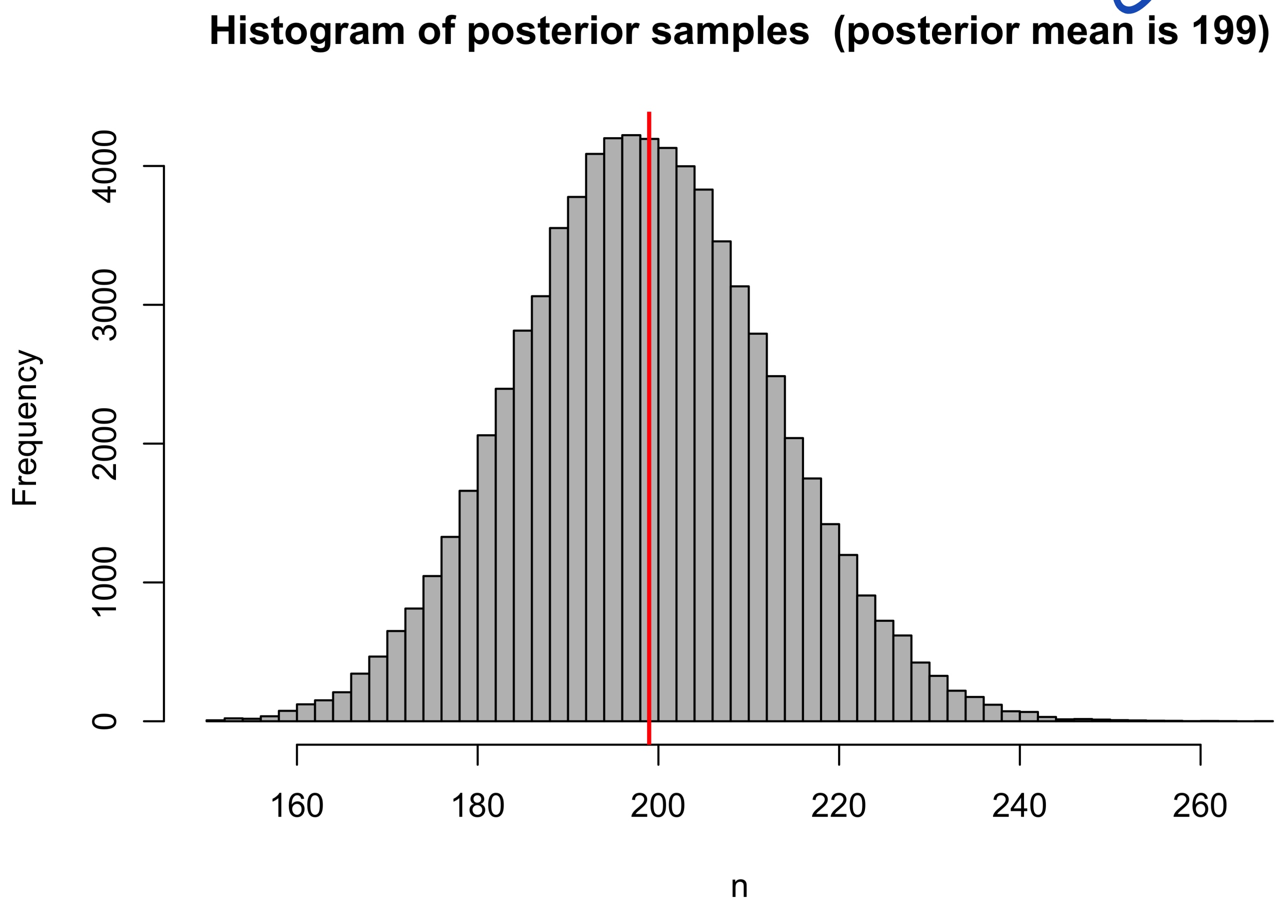
Acceptance rate: 0.48257

The German Tank Problem, Z = 200



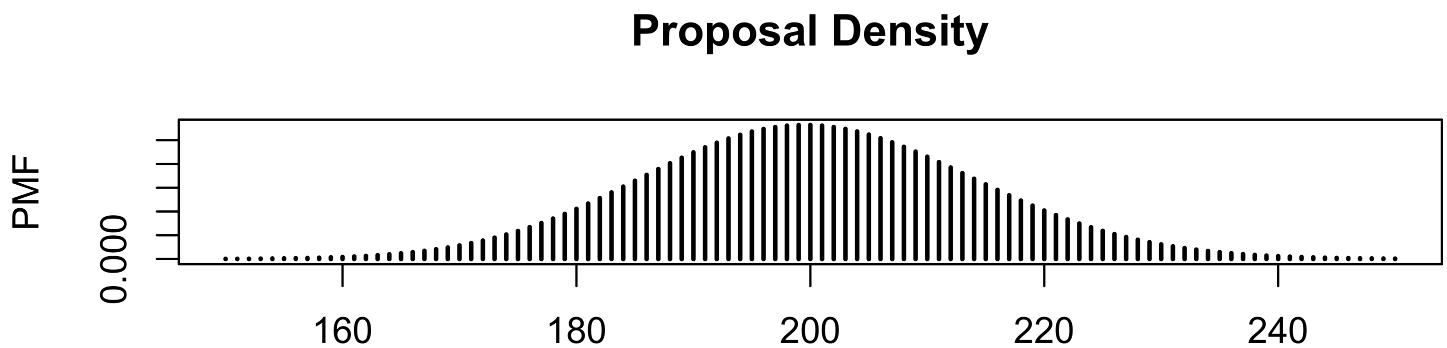


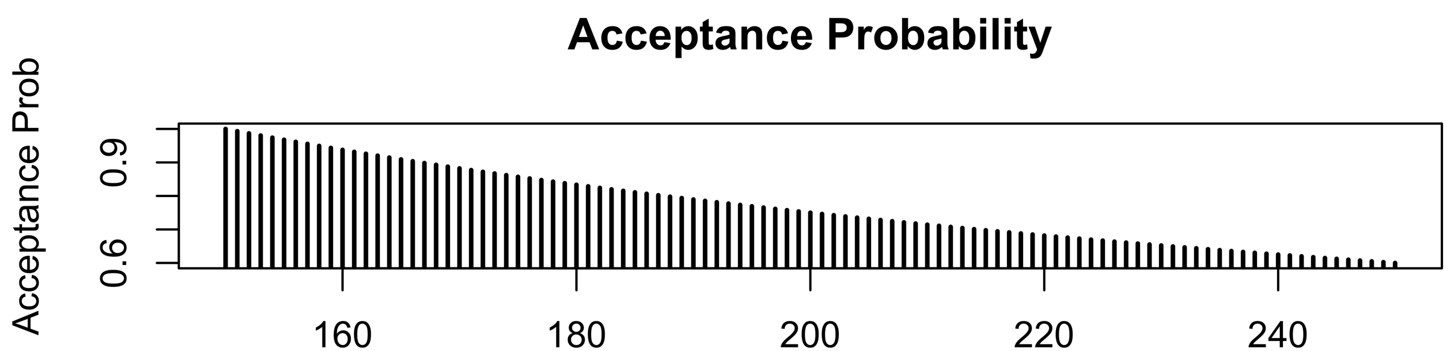
The German Tank Problem, ~~Z~~ y = 150



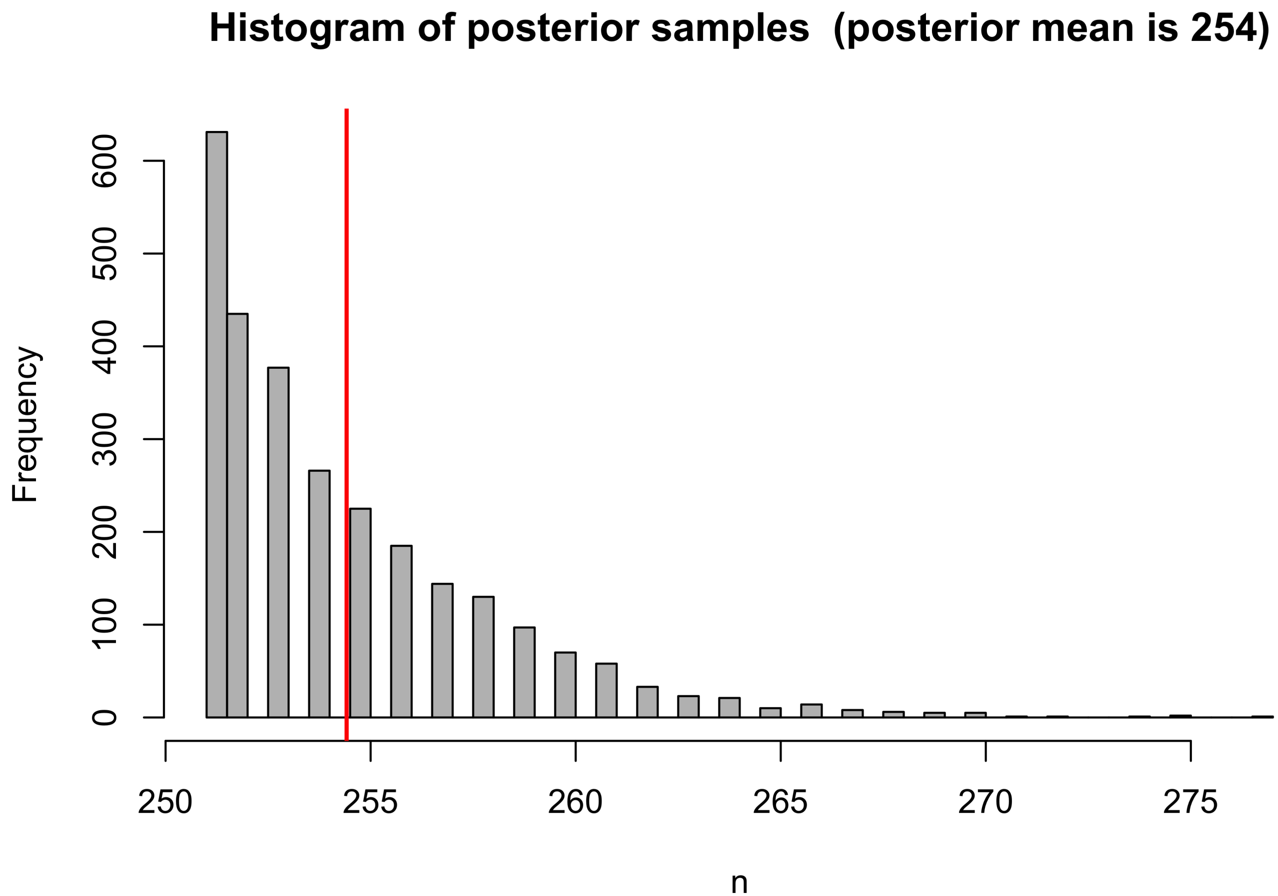
Acceptance rate: 0.75292

German Tank Problem, Z=150





The German Tank Problem, Z = 250



Acceptance rate: 2.749^{-4}

The German Tank Problem, $\cancel{Z}\ y = 250$